UniDPP

DIGITAL PRODUCT PASSPORT · MULTI-BODY STEWARDSHIP

UNIDPP CONTRIBUTION

ISO/IEC JTC 5 · DIGITAL PRODUCT PASSPORT

The Digital Product Passport that works at every border — in two pages

2026-11-12
UniDPP — convened under CalConnect, multi-body stewardship

01 · THE CLAIM

The one sentence

The proposal can be stated in travel-document terms: **a passport for products** — issued once where legal duty attaches, stamped by many jurisdictions, readable at any border, verifiable when the network is not. Jurisdictions are today each inventing their own product passport; what is proposed is the one that works at *every* border — the settlement ICAO achieved for human travel documents (Doc 9303), restated for goods.

> THE PROPOSAL, IN ONE SENTENCE
>
> *A product gets one identity and one immutable event log; every jurisdiction looks at it through its own registered, signed lens and verifies the result under its own trust anchors — so a Japanese inspector, a Chinese customs officer and an EU market-surveillance agent each see their own legally correct view of the same object, offline if necessary, and none of them had to adopt anyone else's rules.*

One thing may lawfully carry several sovereign passports — a CCC mark beside a CE mark on one nameplate — and the framework now says so normatively: same-subject correlation (clause 6.3 k) records the claim, states the counterpart on every read of either side, and never consolidates the two records into one, because each passport answers to its own jurisdiction and its own law. Correlation is the algebra this paper asked for, made explicit.

02 · THE PRECEDENT

The ePassport architecture

No state adopts another state's national identity document, yet all states verify each other's travel documents daily. ICAO Doc 9303 achieves this with a short list of load-bearing elements: one document per person; inspection by each state under its own rules; a master list of trust anchors (the CSCA/PKD system) that makes foreign anchors discoverable without making them mandatory; and a machine-readable zone that supports offline checks. The proposed framework transfers this architecture to products, element for element.

Two consequences follow.

- **A regulation becomes a profile.** A profile is a versioned, registered, signed object that binds data points, transformations, cryptographic suites, trust anchors, access rules and presentation — regulation as data, not regulation as envelope.
- **The symmetry is already conceded inside the EU.** EN 18223 requires a single passport to contain the data of all applicable EU regulations; the proposal generalizes that arrangement symmetrically to the international case, so that no region's envelope is the universal format.

Because cryptographic suites are bound to profiles rather than to the platform, the same payload can carry parallel signatures — SM2/SM3/SM4 where Chinese policy requires them, FIPS suites for United States federal use, eIDAS-acceptable suites in the EU, and ML-DSA for the post-quantum migration — and each verifier checks only what its own acceptance policy requires.

Travel document	Mechanism in the framework
The passport itself	The passport object, issued where legal duty attaches.
Machine-readable zone (MRZ)	Tier-A carrier minimum: offline-verifiable.
Visas on a national document	Lens-scoped attestations by other authorities.
Entry and exit stamps	Verifier custody events.
International Driving Permit	National document with convention renditions = profiles.
CSCA master list (ICAO PKD)	M-of-K trust master list: foreign anchors discoverable, never mandatory.

The border queue with no network	Offline Tier-A verification with cached anchors and as-of verdicts.

03 · THE WORKED SCENE

The border moment

Consider a concrete object from the worked example that threads the reference implementation: the Momiji Mobility E8, an electric bicycle assembled in Kyoto — a drive unit from Rhine Drives (Germany), a battery pack from Weilian (Shenzhen) built of cells from the Haichuan lot — placed on the EU market by a Düsseldorf importer. It arrives at a customs and verification terminal during an ordinary offline window: the Shenzhen host is unreachable, and the EU registry is mid-outage. The officer scans the frame's QR carrier or taps NFC.

What the carrier resolves to is the Tier-A minimum — the machine-readable-zone analogue: identity, status, critical flags and validity, sealed by multi-suite compressed signatures: the Japanese suite, an EU eIDAS-acceptable ECDSA signature and a post-quantum ML-DSA co-signature. The resolver's cached linkset points at the national mirror, and an inclusion proof from the mirror's copy of the transparency log substitutes for the unreachable origin. The terminal returns a verdict of PASS, stamped as of the moment of verification, with a coverage report stating exactly what was not reachable and naming the reading it answers — cryptographic, evidentiary or current-state.

An envelope implementation offers two outcomes at this border: an error page, or a green page that silently trusted whatever it could reach. Here degradation is explicit. The coverage report is the honesty mechanism, and the officer's decision remains auditable as of the moment it was made.

The same system holds where no regulation reaches. A 1962 Super Cub with no living manufacturer and no connectivity carries identity and provenance voluntarily, commissioned by an auction house as qualified attestor — for such objects, provenance *is* the product.

04 · THE SUMMARY

Five points for the plenary

1. **The border test.** Any inspector, in any jurisdiction, offline, obtains a verdict with a coverage report. This is the property that separates a passport system from a lookup service, and it is demonstrable.
2. **Your rules are first-class.** A profile is a signed, versioned, registered object: regulation as data. Jurisdictional authority is preserved by construction, not by exception.
3. **Nothing is invented.** Every component is an existing standard under multi-body stewardship — ISO/IEC 15459 identification, ISO 8601 time, ISO 19135 registration, ISO 80000 quantities, ISO/IEC 18004 carriers, the SIGNATIF trust framework. The work is assembly; the EN series' own technology-neutrality language is the citation.
4. **The load-bearing gaps are named.** §4.3 of EN 18223 leaves the registry vacant; the EN series defines no event model, no state machine and no trust mechanics. The proposed new work items fill exactly those gaps, on the ENs' own terms.
5. **It runs today.** The reference implementation, the conformance evidence and the live services are public and reproducible at unidpp.org. Delegations are invited to register their own profile.

unidpp.org

Reference implementation, conformance evidence, live services, and the worked example of section 03.

Companion papers in this series: 1 framework · 2 semantic registry · 3 lifecycle events · 4 reference implementation · 5 cross-recognition trust · 6 terminology dataset.

UniDPP project, convened under CalConnect — multi-body stewardship; Apache-2.0/MIT per repository.

Paper 7 · dpp-works-at-every-border.html · 2026-09-07